Project Proposal: Healthcare Data Analysis & BI Reporting System

Team Name: **DataX** (Group D)
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1. Project Overview, Objectives, and Scope

1.1 Project Overview

The proposed project aims to develop a comprehensive **Healthcare Data Analysis & BI Reporting System**. To ensure complete compliance with HIPAA/GDPR and zero risk to patient privacy during the development phase, the system will be built and validated using a realistic, **generated (synthetic) dataset containing 100,000 rows**. By leveraging Python, SQL, and Power BI, this platform will provide hospital administrators, clinicians, and financial officers with actionable, historical, and diagnostic insights to improve patient care and operational efficiency.

Figure 1.1 HealthCare

1.2 Objectives

- **Enhance Clinical Quality:** Track patient outcomes, monitor readmission trends, and analyze chronic care distribution across patient demographics.
- **Optimize Financial Health:** Analyze claim denials, visualize supply costs, and track the utilization of high-value medical equipment.
- **Streamline Operations:** Identify bottlenecks in patient flow, monitor bed occupancy trends, and highlight staff workload patterns.
- **Elevate Patient Experience:** Evaluate digital engagement and aggregate patient satisfaction scores alongside wait times.

1.3 Project Scope

In-Scope:

- Generation and ingestion of a synthetic healthcare dataset (100,000 rows of patient records, financial logs, and operational data).
- Data cleaning and transformation using Python and SQL.
- Development of four core interactive Power BI dashboards: Clinical, Financial, Operations, and Patient Experience.
- Publishing and configuring workspaces in Power BI Service for stakeholder access.

Out-of-Scope:

- Integration with live Electronic Health Records (EHR) containing real Protected Health Information (PHI).
- AI, Machine Learning, or predictive modeling (the system is exclusively focused on descriptive and diagnostic data analysis).

2. Project Plan & Resource Allocation

2.1 Timeline & Gantt Chart (4-Week Lifecycle)

Phase	Description	Duration
1. Foundation	Data Generation, Load data into database, Create Star Schema, Make Connection	1 Week
2. Processing	Data cleaning,Feature Engineering, and writing analytical queries.	1 Week
3. Analysis	EDA,Write CTEs (Common Table Expressions**)**, Create SQL Views	1 Week

4. Power BI & Final Documentation	DAX measures, visualizing the four core Power BI dashboards. publishing reports to Power BI Service and Final Documentation.	1 Week

Fig 2.1 Gantt Chart

2.2 Milestones & Deliverables

1. **Milestone 1 (End of Week 1):** Approved Project Charter and delivery of the raw 100,000-row synthetic dataset and SQL Database populated with a verified Star Schema.
2. **Milestone 2 (End of Week 2):** A "Cleaned Master Table" in SQL and a Python validation report.
3. **Milestone 3 (End of Week 3):** A summary of insights, finalized SQL Views for the BI tool and all complex KPI logic (DAX/SQL) is tested and confirmed.
4. **Milestone 4 (End of Week 4):** Interactive Power BI Dashboard, Technical Documentation, and Final Presentation.

2.3 Resource Allocation

Technology Tools:

- **SQL :** 40% of time — Setting up the database and ensuring data integrity.
- **Python :** 30% of time — Handling complex logic, predictions, and cleaning.
- **Power BI :** 30% of time — Designing the UX/UI and writing DAX measures.

3. Risk Assessment & Mitigation Plan

Risk Description	Impact (1-5)	Likelihood (1-5)	Mitigation Strategy
Synthetic Data Lacks Realism	4	3	Collaborate closely with the Healthcare SME during data generation to ensure statistical distributions match real-world hospital data.
Data Model Complexity in Power BI	4	3	Process heavy transformations in SQL views *before* importing to Power BI to keep the BI data model clean and fast.
Scope Creep	4	4	Strictly adhere to the approved project scope. Any new metric requests will be pushed to a Phase 2 update.
Power BI Service Access Issues	3	2	Identify stakeholder accounts and workspace permissions early in Week 3 to ensure smooth deployment in Week 4.

Figure 3.1 Risk Assessment

4. Key Performance Indicators (KPIs) Monitored in Dashboards

These KPIs will be calculated using historical data analysis and displayed across the Power BI workspaces.

4.1 Clinical Analytics (Quality of Care)

- **30-Day Readmission Rate:** Tracking the percentage of patients returning within 30 days, visualized by diagnosis (e.g., Heart Failure, COPD) and demographics.
- **Hospital-Acquired Infection (HAI) Incidence:** Historical rate measured per 1,000 patient days to monitor patient safety.
- **Chronic Disease Distribution:** Demographic breakdown and frequency of high-risk chronic patient admissions.
- **Post-Discharge Follow-Up Compliance:** Tracking the historical percentage of patients who received documented follow-up care within 48 hours of discharge.

4.2 Finance Analytics (Resource & Cost Management)

- **Revenue Analysis by Service Line:** Aggregated revenue mapped by department (e.g., Orthopedics, Cardiology) to highlight financial performance.
- **Insurance Claim Denial Rate:** Tracking the percentage of claims denied, categorized by payer and denial reason codes.
- **Equipment Utilization Rate:** Analysis of logged operational time for expensive assets (MRI/CT scanners) against available hours.
- **Supply Cost per Admission:** Average consumable costs calculated per patient admission over time.

4.3 Operations Analytics (Workflow & Capacity)

- **Average Length of Stay (LOS):** Historical average days admitted, segmented by diagnosis severity.
- **ER Throughput & Wait Times:** Average wait time from emergency room arrival to initial physician assessment.
- **Bed Occupancy Rate Trends:** Daily and weekly tracking of bed utilization to identify historical peak bottleneck periods.
- **Staff Overtime Patterns:** Tracking total overtime hours logged across departments to identify potential staffing shortages or burnout risks.

4.4 Patient Experience & Engagement

- **Patient Satisfaction Score / NPS:** Aggregated average score (0–100) derived from historical post-discharge surveys.
- **Telehealth / Digital Portal Adoption:** Volume of patients utilizing digital tools for

scheduling or consults over the dataset period.

- **Appointment Scheduling Lead Time:** Average days calculated between a patient appointment request and the actual scheduled date.
- **Perceived vs. Actual Wait Time:** Comparative analysis of logged system wait times versus patient-reported wait times via survey data.

5. References

- https://www.statista.com/topics/13277/healthcare-in-egypt/?utm_source=chatgpt.com#topicOverview
- https://www.sciencedirect.com/topics/medicine-and-dentistry/healthcare-research